EE174 Project proposal
Nano quadcopter
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Introduction
For our EE174 final project we want to make a working quadcopter, the project consists of 3 parts, a mechanical, electronics and software part.

A) **Mechanical part**
We plan on building as much of the drone ourselves so we will be 3D printing a frame for the drone, the frame will hold the motors, PCB and batteries in place. It's important to balance out the frame, what we mean by that is that the centre of mass should be in the middle of the quadcopter, this can be done by adding material to the drone.

B) **Electronic part**
The most important part is the electronics part, the brain of the project will be an Arduino nano BLE. We chose this controller because of the build in BLE communication interface + library's and the amount of hardware interfaces (PWM, I^2C…).

A quadcopter consists of 4 motors that can each spin up at a different speed, we bought brushed DC motors that spin up to 49000 RPM at 3.3V. At this speed the motors will need a current of 3A, of course we don't want the motors to be at full speed all the time, to regulate the speed of the motors we will be using PWM to send a pulse signal to the motors.

The Arduino nano has 5 hardware PWM pins of which 4 (1 for each motor) will be connected to a P-channel MOSFET to create the PWM pulse signal. Moreover, we integrated a reverse battery protection system using P-MOSFETs.

What makes a drone complex is the ability to hover steadily in the air it can do this through the help of a Inertial Measurement Unit (IMU). Essentially IMU is used to calculate the angle between the quadcopters each axis and the ground. We chose to use the cheap MPU6050 IMU which includes a 3-axis Gyroscope and a 3-axis Accelerometer.

C) **Software**
The drone won't be a fully fletched drone with a RC remote which offers precise control but rather a proof of concept. We will control the drone over BLE using a terminal on our phone or if time allows us we will make an app in the MIT app inventor workspace.

We will be required to write a program to read voltage of the battery. This will be done by measuring the analog measurement by Arduino a voltage reference. We set the internal reference voltage to 1.1V. Then we set a program to stabilise the motors. This is done by implementing a PID controller. This implementation is done by stabilising the pitch and roll of MPU6050. A function will be created to stabilize motors based on the required speed. Lastly, we are required to program the BLE connectivity from the Arduino Nano to our phones. The BLE is Bluetooth Low Energy that doesn't require too much power to operate. This is done by being delivered bursts of pulses. The nano connects to the BLE chip using the built in Serial on pins TX and RX. Of course this slightly makes it more difficult if you want to talk to the PC instead, however I will rarely need a connection to the PC anyway.